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Project 11

NTP Point-Kinetics & Hydrogen Reactivity Solver

Summary (what, why, and output)

Building upon the cryogenic fluid dynamics mapped in Project 10, this simulation bridges the gap between aerospace fluid mechanics and core nuclear physics. In a Nuclear Thermal Propulsion (NTP) engine, liquid hydrogen acts as a highly effective neutron moderator. If the propellant flow surges and denser, colder hydrogen floods the core, it suddenly slows down more neutrons, inducing a massive positive reactivity spike.

This Python program models that exact dynamic threat. By solving the 6-group delayed neutron point-kinetics equations using a numerical forward Euler method, it simulates a transient engine startup scenario. The program injects a sudden $0.003k/k$ reactivity step (simulating a cold hydrogen surge) into a reactor operating at a baseline 100 MW. The output is a time-history plot demonstrating the terrifying "prompt jump" in reactor power caused by the propellant density shift, proving the necessity of precise, millisecond-level control drum algorithms in space reactors.

Tools and Assumptions

Languages: Python

Libraries: numpy, matplotlib

Kinetics Model: 6-Group Point-Kinetics (Standard U-235 delayed neutron fractions and decay constants)

Prompt Neutron Generation Time (Λ): 1×10^{-4} seconds

Initial State: Steady-state operation at 100 Megawatts (MW)

Transient Trigger: Sudden step insertion of $+0.003$ reactivity at $t = 1.0$ second (simulating a cold liquid hydrogen surge).

Solver: Explicit Forward Euler numerical integration with a 1-millisecond time step ($dt = 0.001$ s).

Engineering Theory

The fundamental challenge of Nuclear Thermal Propulsion is that the rocket's fuel (the nuclear reactor) and the rocket's thrust medium (the hydrogen propellant) are intimately coupled. In a standard reactor, controlling power requires moving mechanical control rods. In an NTP engine, the fluid dynamics of the hydrogen actively change the nuclear physics in real time.

$$\frac{d}{dt}n(t) = \frac{(k - 1)}{l}n(t)$$

$$n(t) = n(0)\exp\left[\frac{(k - 1)}{l}t\right]$$

Figure 1: Point Kinetics 6 group delayed neutrons equations

To model dynamic reactor power, we rely on point-kinetics equations. As shown in Figure 1, the basic differential equation tracks the rate of change of the neutron population, $n(t)$, over time. When reactivity is injected into the core, it alters the neutron multiplication factor (k), driving an exponential response in reactor power. For high-accuracy simulations, this basic equation is expanded into the 6-group delayed neutron point-kinetics model used in our Python solver, which accounts for the delayed release of neutrons from decaying fission products.

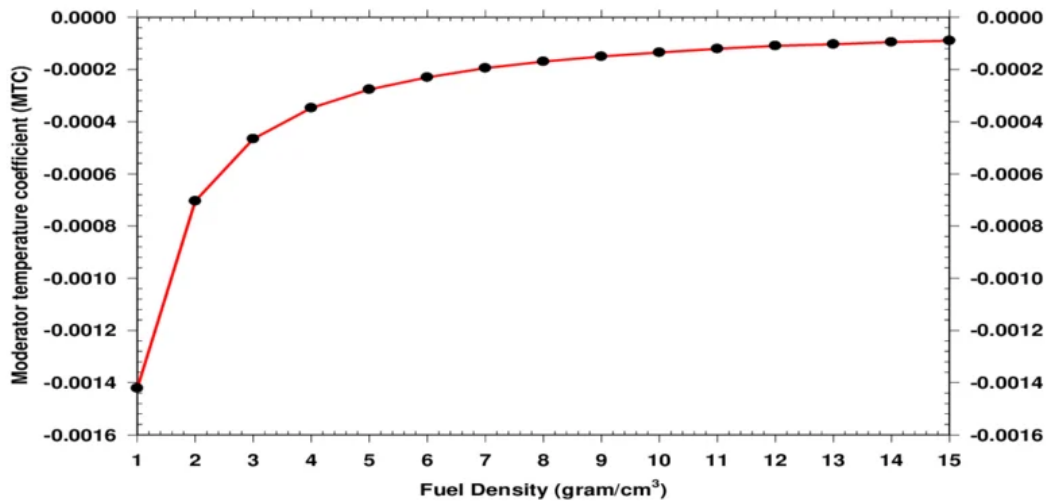


Figure 2: Moderator temperature coefficient diagram

The physical trigger for these power spikes in an NTP engine is reactivity feedback. Figure 2 visualizes how changes in density drastically alter the Moderator Temperature Coefficient (MTC). Hydrogen is a highly effective neutron moderator. If an engine anomaly causes a sudden surge of dense, cryogenic liquid hydrogen to flood the core, it over-moderates the system. This dense fluid slows down too many neutrons at once, which inadvertently injects massive positive reactivity into the core, forcing the exponential power spike governed by the equations in Figure 1. Our simulation specifically models this transient threat to prove why active control drum adjustments are mandatory during an engine burn.

Procedure

The Python script is built around a numerical explicit Forward Euler solver to handle the coupled differential equations of reactor kinetics. We first import numpy and define the standard U-235 nuclear constants: the six decay constants and the six delayed neutron fractions, along with a prompt neutron generation time of 1×10^{-4} seconds.

The simulation environment is set to track 10 seconds of time using a 1-millisecond step size to prevent numerical instability during rapid exponential power changes. The reactor state variables. Power (P) and the six delayed neutron precursor concentrations are initialized assuming perfect steady-state equilibrium at 100 MW.

To simulate the operational threat of an NTP engine, the script introduces a conditional trigger. At exactly $T = 1.0$ second, the reactivity array spikes from 0.0 to +0.003k/k, physically representing a dense slug of cold hydrogen suddenly flushing into the core and over-moderating the neutrons. The for loop calculates the exact rate of change for both the power and the precursors at every millisecond, continuously updating the arrays. Finally, matplotlib plots the resulting power excursion against time to visualize the transient response.

Results

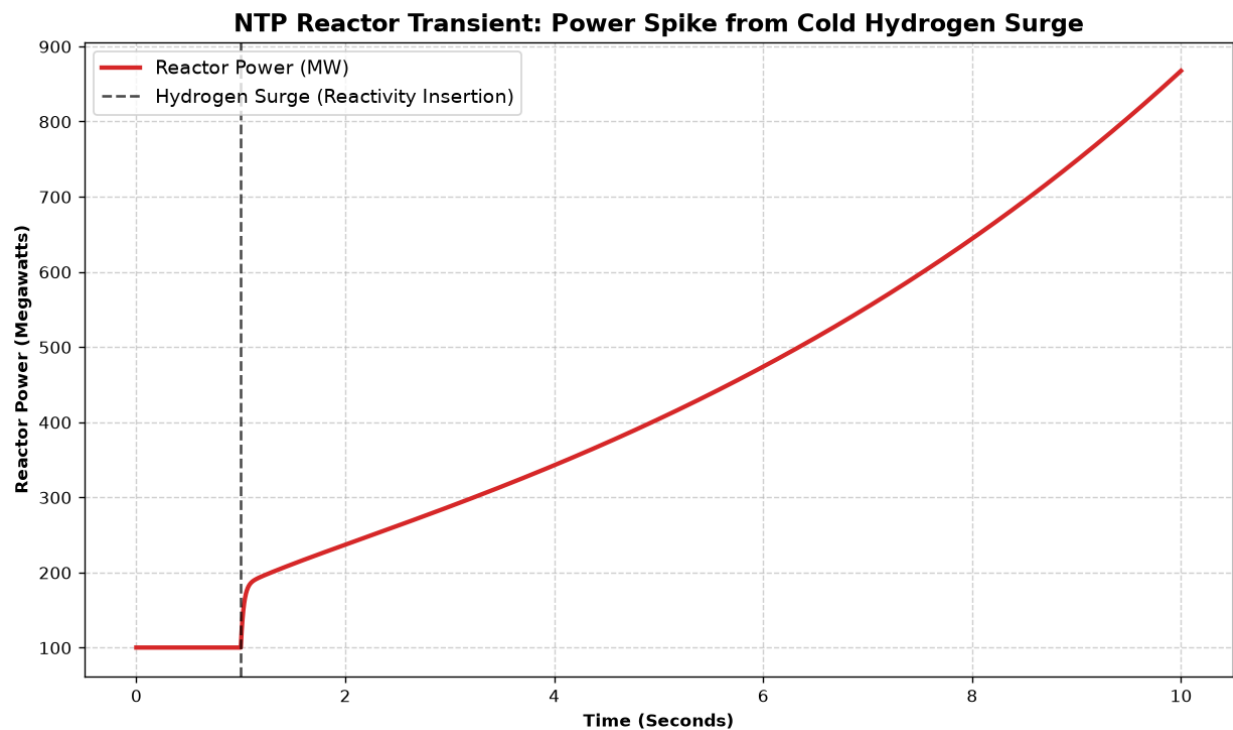


Figure 3: Power Spike from Cold Hydrogen Surge

The explicit Forward Euler solver successfully captured the terrifying reality of reactor transient physics, as visualized in the time-history plot. For the first second of the simulation, the engine is humming along in perfect equilibrium at a steady 100 MW. At exactly $t = 1$ second, the vertical dashed line marks our simulated anomaly: a sudden surge of dense, cold liquid hydrogen flooding the core.

The immediate physical response is a classic nuclear "prompt jump." Because the extra mass of hydrogen instantly slows down a massive amount of neutrons, the reactor power nearly doubles in a fraction of a millisecond, rocketing almost vertically to roughly 200 MW. Following this initial violent spike, the delayed neutrons begin to take over, driving the power into a steep exponential climb that approaches a catastrophic 900 MW by the end of the 10-second window.

This graph perfectly illustrates why autonomous control systems are the hardest engineering challenge in Nuclear Thermal Propulsion. If a propellant surge occurs in space, the flight computer has to recognize the fluid density shift and physically rotate the neutron-absorbing control drums within milliseconds to catch that prompt jump, otherwise the exponential power surge will melt the reactor.

Future Scope

With the baseline thermodynamic equations (Project 9), the internal fluid phase changes (Project 10), and the dynamic nuclear reactivity (Project 11) successfully modeled in Python, the theoretical foundation of this Nuclear Thermal Propulsion portfolio is complete.

The next and final phase scales this 1D computational logic into fully meshed 3D environments. Project 12.1: 3D Supersonic De Laval Nozzle CFD will transition into SolidWorks and ANSYS Fluent. By activating the compressible flow and energy equation models, we will physically design a 24:1 expansion nozzle, apply our 2500 K hydrogen operating parameters, and visually map the Mach 3+ supersonic expansion, shock diamonds, and turbulent boundary layers of a functioning NTP engine.

References

1. Duderstadt, J. J., & Hamilton, L. J. (1976). Nuclear Reactor Analysis. John Wiley & Sons.
2. Taube, M. (1974). Plutonium - A General Survey.